Hugo Ruivo

Senior Software Engineer | Real-Time Systems & Simulation

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Professional Summary

Software engineer and technical lead with 10+ years of experience in real-time 3D, simulation, and automotive visualization. Builds C++ and Unreal Engine applications, modular software architectures, and production tooling. Leads developers and artists from R&D through delivery. Personal projects explore flight automation, compilers, systems programming, and FPGA design.

Core Skills

Languages: C, C++, Python, Zig, Rust, C#, TypeScript, and VHDL through personal projects.

Real-time development: Unreal Engine, Unity, physics simulation, interactive 3D local and streamed applications, performance profiling and optimization.

Architecture and tooling: modular and data-driven systems, web integrations, rendering pipelines, Jenkins CI/CD, Perforce, Git, Deadline, Houdini, Blender.

Leadership: technical direction, prototyping, cross-functional teams, software delivery.

Professional Experience

Real-Time R&D Lead | Digital Giant

May 2022 - Present

- Lead software architecture, performance, tooling, and delivery for automotive visualization and configurator projects.
- Architect a modular, data-driven Unreal Engine framework supporting multiple vehicle brands, rendering workflows, and interactive products.
- Established a scalable Perforce structure for shared vehicle data, reusable plugins, and project delivery.
- Integrated Deadline with cloud rendering workers and built Jenkins CI/CD pipelines using Python, shell scripts, and Unreal Engine automation tests.

CGI Real-Time R&D Lead | Hogarth Worldwide

2020 - 2022

- Led development of an Unreal Engine vehicle configurator with multiple environments and cinematic sequences.
- Integrated web-based systems with Unreal Engine for product validation and stakeholder feedback.

Technical Artist | Burrows

2019 - 2020

- Created Unreal Engine tools for vehicle import, configuration, and real-time rendering workflows.
- Built a WebGL vehicle configurator with an automated Houdini pipeline to optimize complex assets for browser rendering.

Software Developer | Mackevision

2018 - 2019

- Maintained Unreal Engine vehicle configurators using Pixel Streaming and extended interactive functionality.
- Implemented communication between an iPad control interface and the main Unreal Engine application.

Software Developer | Abyssal (now Ocean Infinity)

2016 - 2018

- Developed an Unreal Engine remotely operated vehicle simulator for the oil and gas industry.
- Implemented buoyancy, tether physics, hydraulic manipulator simulation, multi-display rendering, and custom hardware input integration.
- Designed modular vehicle systems, data-driven missions, and session recording; led developers and artists as the prototype expanded.

Game Programmer | X-Team

2012 - 2016

- Developed web and mobile game projects using Unity.

Selected Personal Projects

DCS Autonomous Aircraft

Flight firmware using Lunara for autonomous takeoff, waypoint navigation, and landing of the F-16 and Su-25 in DCS. Telion streams telemetry to Mission Dashboard and relays commands to the aircraft.

Lunara Language and Compiler

Zig compiler targeting bare-metal RISC-V (RV32I), inspired by NASA/JPL's Power of Ten rules. Includes unit-aware types, checked arithmetic, mandatory wait timeouts, fault handling, mission scenarios, and fault injection.

Telemetry and Mission Control

Telion telemetry backend and a web dashboard for monitoring KSP/DCS vehicles and sending commands. Built an adapter to display NASA Artemis II telemetry using the same data format and dashboard.

CHIP-8 FPGA Core

CHIP-8 implemented in VHDL, running on the GameBub console's FPGA.

AI Assisted Car Configurator

Integrated AI agents with an Unreal Engine vehicle configurator using Python, C++, and OpenAI/llama models.

Education

Game Development Engineering | Polytechnic Institute of Cavado and Ave

Barcelos, Portugal | 2009 - 2012